

QUIZ 1: PROBABILITY 3

Answer all questions; each question carries 5 marks.

Problem 1.

1. Define: sigma algebra and a probability space.
2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\Omega_0 \in \mathcal{F}$ with $\mathbb{P}(\Omega_0) > 0$. Show that $(\Omega_0, \mathcal{F}_0, \mathbb{P}_0)$ is a probability space, where

$$\mathcal{F}_0 := \{A \cap \Omega_0 : A \in \mathcal{F}\} \quad \& \quad \mathbb{P}_0(A) := \frac{\mathbb{P}(A)}{\mathbb{P}(\Omega_0)}, \quad \forall A \in \mathcal{F}_0.$$

Problem 2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Prove that there exist at most countably many pairwise disjoint measurable sets of positive measure.

Problem 3. Let $(S_n)_{n \geq 0}$ denote the simple random walk (SRW) on $\mathbb{Z}$ starting at 0, i.e., $S_0 = 0$. Answer the following questions:

1. What is the distribution of the random vector (S_1, S_3) ?
2. Determine the probability that $S_1 = S_3$.

Problem 4. Determine if the following statement is true or false, along with justification:

Let $\mathcal{F} = \{(a_s, b_s)\}_{s \in \mathcal{S}}$ be a collection of open intervals in $[0, 1]$ such that the set of end points of these intervals is dense in $[0, 1]$. Let $\sigma(\mathcal{F})$ and $\mathcal{B}([0, 1])$ be the sigma algebra generated by $\mathcal{F}$ and the Borel sigma algebra on $[0, 1]$ respectively. Then $\sigma(\mathcal{F}) = \mathcal{B}([0, 1])$.